PROJECT TITLE

Adaptive Path Planning for Autonomous Warehouse Robots Using Deep Reinforcement Learning

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BACHELOR OF ENGINEERING / BACHELOR OF TECHNOLOGY

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BONAFIDE CERTIFICATE

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DECLARATION

I , M.Nandhini (192472318) of the Computer Science and Engineering, SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled "Adaptive Path Planning for Autonomous Warehouse Robots Using Deep Reinforcement Learning" is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

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Student	Responsibilities	Design / Development	Testing / Analysis	Report	Contribution
M. Nandhini	Responsible for **Module 2 – DQN-Based Adaptive Path Planning**	Designed and developed the DQN-based path-planning module, action-selection process, reward and penalty mechanism, Experience Replay, and Target Network components	Tested the DQN navigation process and analysed path efficiency, collision avoidance, success rate, completion time, and reward	Prepared the Module 2 design, implementation, testing, results, conclusion, references, and appendix sections	**50%**
K. L. Khaviya Varshini	Responsible for **Module 1 – Warehouse Environment and State Management**	Designed and developed the warehouse environment, robot and destination representation, obstacle and shelf representation, state management, and robot movement actions	Tested the warehouse environment, robot movement, state updates, obstacle detection, and environment responses	Prepared the Module 1 design, implementation, testing, results, and related report sections	**50%**

ABSTRACT

Modern warehouses increasingly use Autonomous Mobile Robots (AMRs) to transport goods between storage, picking, and delivery locations. Efficient path planning is essential for reducing travel distance, completion time, energy consumption, and the possibility of collisions. Traditional path-planning techniques such as A*, Dijkstra's algorithm, potential field methods, and rule-based navigation generally depend on predefined maps and fixed navigation strategies. These approaches may become less effective when the warehouse environment changes or when unexpected obstacles are introduced.

This project proposes an Adaptive Path Planning System for Autonomous Warehouse Robots using Deep Reinforcement Learning (DRL). The proposed system models the warehouse as a two-dimensional environment containing a robot, destination, shelves, static obstacles, and dynamic obstacles. The robot observes its current state, including its position, target location, and surrounding obstacles, and selects an appropriate movement action. A Deep Q-Network (DQN) is used to learn the value of possible actions and gradually develop an efficient navigation policy through repeated interaction with the environment.

A reward-based learning mechanism is used to encourage successful destination arrival and efficient movement while penalizing collisions, unnecessary movements, and longer paths. Experience Replay and a Target Network are incorporated to improve the stability of the DQN training process. The performance of the proposed system can be evaluated using parameters such as success rate, path length, collision rate, completion time, and accumulated reward. The proposed approach is also intended to be compared with a conventional path-planning technique such as A*.

The project aims to provide an adaptive, collision-free, and efficient path-planning solution that can respond to changing warehouse conditions. The expected outcome is a trained DQN-based navigation model capable of learning improved paths from a source location to a destination while minimizing unnecessary movement and collisions. The system can further be extended to support multiple robots, real sensor data, dynamic warehouse environments, and real-world robotic platforms.

Keywords: Deep Reinforcement Learning, Deep Q-Network, Autonomous Mobile Robot, Path Planning, Warehouse Navigation, Obstacle Avoidance

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LIST OF ABBREVIATIONS / SYMBOLS

Abbreviation / Symbol	Description	Unit
IoT	Internet of Things	—
RTC	Real-Time Clock	—
MCU	Microcontroller Unit	—
Wi-Fi	Wireless Local Area Network connectivity	—
OLED	Organic Light-Emitting Diode display	—
PWM	Pulse Width Modulation	—
BPM	Beats Per Minute	bpm
°C	Degree Celsius	°C
V	Voltage	V
I	Current	mA
t	Time	s / min

CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

Modern warehouses increasingly use Autonomous Mobile Robots (AMRs) to transport goods between storage areas, picking locations, and delivery points. Efficient robot navigation is an important requirement because the selected path directly affects travel distance, execution time, energy consumption, and the possibility of collisions.

Traditional path-planning methods such as A*, Dijkstra's algorithm, and rule-based navigation are commonly used for robot movement. These approaches generally depend on predefined maps, fixed rules, or manually designed navigation strategies. Although they can provide effective paths in structured environments, their performance may be affected when the warehouse layout changes or when unexpected obstacles appear.

Deep Reinforcement Learning (DRL) provides an alternative approach in which a robot learns navigation decisions through interaction with its environment. The robot observes its current state, selects an action, receives a reward or penalty, and gradually improves its navigation policy. In this project, a Deep Q-Network (DQN) is used to develop an adaptive path-planning approach for autonomous warehouse robots. The system aims to enable efficient, adaptive, and collision-free navigation from a source location to a destination.

1.2 Need for the Project

Efficient path planning is necessary for improving the operational performance of autonomous warehouse robots. Warehouse environments may contain shelves, storage areas, robots, and other obstacles that can affect navigation. A robot must therefore identify a suitable path while avoiding obstacles and reaching its destination efficiently. Existing path-planning techniques often require a predefined environment or repeated path replanning when significant changes occur. Rule-based systems may also have difficulty handling all possible navigation situations because their behaviour depends on predefined movement and obstacle-avoidance rules.

Therefore, there is a need for an adaptive navigation system that can learn from interaction with the warehouse environment. Deep Reinforcement Learning can support this requirement by enabling the robot to improve its movement decisions based on rewards and penalties. The proposed project addresses

the need for a navigation approach that aims to reduce path length, travel time, unnecessary movement, and robot-obstacle collisions while improving adaptability to changing warehouse conditions.

1.3 Problem Statement

Autonomous warehouse robots must navigate efficiently from a source location to a destination while avoiding shelves and obstacles. The engineering problem involves multiple interacting factors, including robot position, destination location, surrounding obstacles, movement decisions, path length, travel time, and collision avoidance. Traditional navigation approaches such as A*, Dijkstra's algorithm, potential field methods, and rule-based navigation can provide paths based on predefined maps or rules. However, these methods have limitations in adapting to unexpected obstacles and changing warehouse conditions. Frequent environmental changes may require path replanning, while rule-based systems cannot easily handle every possible situation.

The problem is therefore to develop an adaptive path-planning system using Deep Reinforcement Learning that enables an autonomous warehouse robot to learn efficient movement decisions, avoid obstacles, reduce unnecessary movement, and successfully reach its destination.

1.4 Project Aim

The aim of this project is to develop an adaptive path-planning system for autonomous warehouse robots using Deep Reinforcement Learning and a Deep Q-Network (DQN) to enable efficient and collision-free navigation in a changing warehouse environment.

1.5 Project Objectives

The objectives of the project are:

1. To develop a warehouse environment containing a robot, destination, shelves, and obstacles.
2. To represent the robot's state using its current position, target location, and nearby obstacles.
3. To define suitable navigation actions such as Up, Down, Left, Right, and Stop.
4. To develop a DQN-based agent for selecting appropriate robot movements.
5. To use a reward-based mechanism that rewards efficient navigation and destination arrival.
6. To penalize collisions, unnecessary movements, and longer paths.

7. To reduce the probability of robot-obstacle collisions.
8. To minimize path length and travel time.
9. To improve navigation performance when the warehouse environment changes.
10. To evaluate the system using success rate, path length, collision rate, completion time, and reward.
11. To compare the proposed DRL-based approach with a traditional path-planning method such as A*.

1.6 Scope

Included Scope

The project focuses on the development of a simulated warehouse path-planning environment. The warehouse environment includes a robot, destination, shelves, and obstacles. The robot observes the current environment and selects navigation actions to move towards the target location. The main scope includes the use of a Deep Q-Network (DQN) for learning navigation decisions. The system uses rewards and penalties to improve the robot's behaviour over multiple training episodes. Experience Replay and a Target Network are also included to improve the stability of the training process.

The system performance is intended to be evaluated using metrics such as success rate, path length, collision rate, completion time, and accumulated reward.

Excluded Scope

The current project scope does not include a complete real-world warehouse deployment. Integration with physical robots, advanced real-time sensor systems, and large-scale multi-robot coordination are outside the primary implementation scope.

Assumptions

The warehouse is represented as a two-dimensional environment. The robot can obtain the required state information from the environment, including its position, target location, and information about surrounding obstacles.

Boundary Conditions

The robot operates within the defined warehouse environment and selects actions from the available movement set. Collisions and successful destination arrival are detected by the environment and corresponding feedback is provided to the DQN agent.

1.7 Expected Engineering Outcome

The expected engineering outcome of this project is a DQN-based adaptive path-planning system for an autonomous warehouse robot.

The developed system is expected to:

- Create and manage a two-dimensional warehouse navigation environment.
- Represent the robot's current navigation state.
- Detect obstacles, collisions, and destination arrival.
- Select appropriate robot movements using a trained DQN agent.
- Learn improved navigation behaviour through rewards and penalties.
- Generate efficient and adaptive paths towards the destination.
- Reduce unnecessary movements and collisions.
- Evaluate navigation performance using success rate, path length, collision rate, completion time, and reward.

The final outcome is intended to demonstrate how Deep Reinforcement Learning can provide improved adaptability compared with conventional fixed path-planning approaches. The system also provides a foundation for future extensions involving dynamic obstacles, multiple robots, real sensor data, and real-world robotic platforms.

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

The literature review was conducted to understand the existing approaches used for mobile robot navigation and autonomous path planning. Relevant research papers were selected based on their focus on Deep Reinforcement Learning (DRL), Deep Q-Networks (DQN), mobile robot navigation, multi-robot path planning, and conventional path-planning techniques.

The reviewed studies include research on DRL-based navigation, DQN-based robot path planning, policy-based reinforcement learning, and traditional approaches such as A* and heuristic path-planning methods. The selected literature was analysed based on the techniques used, the engineering problem addressed, and its relevance to the proposed adaptive warehouse robot navigation system.

2.2 Review of Existing Work

2.2.1 Deep Reinforcement Learning Based Mobile Robot Navigation

Zhu and Zhang (2021) reviewed the application of Deep Reinforcement Learning in mobile robot navigation. The study discussed approaches including Deep Q-Networks and policy-based reinforcement learning for improving robot navigation in complex environments. The work highlights the limitations of conventional navigation approaches and the potential of learning-based methods for autonomous decision-making. The study is relevant to the proposed project because it supports the use of DRL for enabling robots to learn navigation policies through interaction with their environment.

2.2.2 Multi-Robot Path Planning Based on DQN

Yang et al. (2020) proposed a multi-robot path-planning approach based on Deep Reinforcement Learning and the Deep Q-Network algorithm. The approach uses DQN, Q-learning concepts, and Experience Replay to support efficient path planning and coordination among multiple robots.

The study demonstrates the applicability of DQN-based learning for robot navigation and path-planning problems. Experience Replay is particularly relevant to the proposed project because it can improve the stability of the learning process by using previously collected navigation experiences during training.

2.2.3 Review of Deep Reinforcement Learning Algorithms for Mobile Robot Path Planning

Singh, Ren, and Lin (2023) reviewed different Deep Reinforcement Learning algorithms used for mobile robot path planning. The techniques discussed include DQN, policy-gradient methods, and actor-critic approaches. The study focuses on the application of DRL techniques to path-planning problems in complex and dynamic environments. This work is relevant because it shows that different DRL algorithms can be used for robot navigation. For the proposed system, DQN is selected as the primary learning method to enable the robot to estimate the value of available actions and gradually learn an effective navigation policy.

2.2.4 Traditional Path-Planning Techniques

Liu et al. (2023) reviewed path-planning techniques used for mobile robots, including A*, heuristic methods, and global and local planning approaches. These techniques are commonly used to identify paths that aim to achieve safe, short, and time-efficient navigation.

Traditional methods can provide effective solutions when the environment is known and relatively stable. However, changing warehouse conditions and unexpected obstacles may require repeated replanning or additional navigation strategies. This limitation motivates the use of adaptive learning-based approaches in the proposed project.

2.2.5 Deep Reinforcement Learning for Adaptive Navigation

Li (2023) reviewed mobile robot navigation approaches based on Deep Reinforcement Learning and Deep Neural Networks. The study focuses on adaptive navigation using learned policies instead of fixed navigation rules.

This approach is closely related to the proposed project because the warehouse robot is expected to learn movement decisions based on its state and environmental feedback rather than relying only on manually defined navigation rules.

2.3 Comparative Analysis

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
Traditional A* Path Planning	Finds efficient paths using heuristic search	Depends on environmental information and may require replanning when conditions change	Used as a conventional approach for comparison

Dijkstra's Algorithm	Provides shortest-path solutions through graph traversal	Can be computationally demanding and does not inherently adapt through learning	Represents a traditional navigation approach
Rule-Based Navigation	Simple and easy to implement	Cannot easily handle all possible situations or unexpected changes	Highlights the need for learning-based navigation
DQN-Based Path Planning	Learns action values and improves navigation through experience	Requires training and appropriate reward design	Selected as the main approach in the proposed system
Policy-Based / Actor-Critic DRL	Supports learning-based decision-making in complex environments	May involve greater implementation complexity	Considered as alternative DRL approaches

Relevance to Proposed Work

- Used as a conventional approach for comparison
- Represents a traditional navigation approach
- Highlights the need for learning-based navigation
- Selected as the main approach in the proposed system
- Considered as alternative DRL approaches

The comparative analysis shows that traditional approaches are useful for predefined and structured environments, whereas DRL-based approaches provide the ability to learn navigation behaviour through interaction. The proposed system therefore focuses on DQN-based adaptive path planning while considering A* as a traditional method for comparison.

2.4 Research / Engineering Gap

The reviewed existing approaches show that conventional path-planning algorithms can effectively identify paths in predefined environments. However, these methods generally depend on fixed maps, manually designed rules, or repeated path replanning when the environment changes.

Warehouse environments can contain obstacles and changing conditions that require robots to make adaptive movement decisions. Rule-based navigation may not effectively handle all possible situations, while conventional algorithms may require updated environmental information before generating a new path. The identified engineering gap is therefore the need for a system that enables an autonomous warehouse robot to learn and improve its navigation decisions through interaction with the environment. The system should consider the robot's position, destination, and surrounding obstacles while reducing unnecessary movement and collisions.

2.5 Proposed Contribution

The proposed project contributes an adaptive warehouse robot path-planning system based on Deep Reinforcement Learning. The system uses a two-dimensional warehouse environment containing the robot, destination, shelves, and obstacles. A DQN agent is used to estimate the value of possible movement actions. The robot receives rewards for efficient movement and successful destination arrival, while penalties are provided for collisions, unnecessary movements, and longer paths. The training process uses Experience Replay and a Target Network to support stable learning.

The proposed contribution is to develop a navigation system capable of learning an adaptive and collision-free path towards a destination. The system is intended to improve navigation performance using evaluation parameters such as success rate, path length, collision rate, completion time, and reward. It also provides a foundation for future development involving dynamic obstacles, multiple robots, real sensor data, and real-world robotic platforms.

2.4 Research / Engineering Gap

From the literature review, it is observed that existing systems commonly focus on one or more of the following functions:

- Generating medicine reminders.
- Providing alarm or notification alerts.
- Monitoring patient health conditions.
- Using smart pill box hardware.
- Providing cloud-based healthcare monitoring.

However, there is a need for a simple and integrated mechanism that specifically focuses on recording medicine intake status and updating this information through a cloud-based platform for remote monitoring

2.5 Proposed Contribution

he proposed Medication Compliance and Cloud Tracking Unit (Module 3) contributes to the overall IoT-enabled Smart Medicine Reminder System by focusing on the confirmation and tracking of medication intake.The main contribution of the module is the development of a mechanism that records whether the patient has taken the medicine and maintains this information using cloud technology. The updated medication status can support remote monitoring and provide caregivers with information about the patient's medication compliance.

The proposed contribution can be summarized as follows:

- Records the patient's medicine intake status.
- Supports medication compliance monitoring.
- Maintains medication information digitally.
- Updates medication status through cloud connectivity.
- Enables remote access to medication tracking information.
- Supports caregivers in monitoring medicine compliance.
- Integrates with the overall smart medicine reminder system.

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

3.1.1 Stakeholder / User Requirements

The proposed system is intended for warehouse environments where autonomous robots are required to move efficiently from a source location to a destination. The primary requirement is to provide an adaptive path-planning mechanism capable of considering the robot position, destination, and surrounding obstacles.

3.1 Requirements

Stakeholder / User	Requirement	Requirement Type	Measurable Target
Warehouse Operator	Robot should reach the assigned destination efficiently	Functional	Successfully reaches the destination
Warehouse Management	Reduce unnecessary travel and improve navigation efficiency	Non-Functional	Reduced path length and completion time
Autonomous Warehouse Robot	Detect surrounding obstacles during navigation	Functional	Obstacles are identified during movement
Autonomous Warehouse Robot	Avoid collisions with shelves and obstacles	Functional	Reduced collision rate
System / DQN Agent	Select appropriate navigation actions	Functional	Select from Up, Down, Left, Right, and Stop actions
System / DQN Agent	Learn efficient navigation behaviour through interaction	Functional	Improved performance over multiple training episodes
Project Developer	Use Deep Q-Network for adaptive path planning	Functional	DQN generates action values and selects the best action
System	Provide rewards and penalties based on robot actions	Functional	Positive reward for efficient movement and goal arrival; penalty for collisions and unnecessary movements

System	Maintain stable learning during training	Non-Functional	Experience Replay and Target Network are used
Project Evaluator	Evaluate the navigation performance	Non-Functional	Success rate, path length, collision rate, completion time, and reward are measured
Future System Users	Support changing warehouse conditions	Non-Functional	Robot can adapt its navigation policy based on environmental interaction

3.2 Constraints & Applicable Standards

The proposed DQN-based adaptive path-planning system is developed within a simulated two-dimensional warehouse environment. The main constraints include the representation of the warehouse, limited robot movement actions, collision avoidance, path efficiency, and the requirement for repeated training. The robot operates using predefined actions such as Up, Down, Left, Right, and Stop, while the environment contains the robot, destination, shelves, and obstacles. The system must minimise unnecessary movement, longer paths, and collisions while enabling the DQN agent to learn improved navigation decisions through repeated interaction with the environment. Experience Replay and a Target Network are used to support stable learning during the training process. Since the current project is simulation-based, real-world robot hardware, sensors, and physical warehouse deployment are outside the present scope and can be considered for future development.

Regarding applicable standards, the project source provided does not specify any particular IEEE, ISO, robotics, or software engineering standard followed during implementation. Therefore, no unsupported standard is claimed in this report. The system is instead designed and evaluated according to the project requirements, including efficient navigation, obstacle avoidance, collision reduction, path length, success rate, completion time, and reward-based performance evaluation. This follows the new report format requirement that only standards and constraints that actually apply to the project should be included.

3.3 Design Specifications

Parameter	Target / Requirement	Unit	Priority
Warehouse Environment	Two-dimensional environment containing robot, destination, shelves, and obstacles	Environment	High
Robot State	Includes robot position, target location, and nearby obstacles	State Information	High

Navigation Actions	Up, Down, Left, Right, and Stop	Actions	High
Learning Algorithm	Deep Q-Network (DQN) for adaptive path planning	Algorithm	High
Collision Detection	Detect robot collision with obstacles	Status	High
Destination Detection	Detect successful arrival at the target location	Status	High
Reward Mechanism	Reward efficient movement and successful destination arrival	Reward Value	High
Penalty Mechanism	Penalise collisions, unnecessary movements, and longer paths	Penalty Value	High

3.4 Alternative Solutions & Evaluation

Criterion	**Weight**	***A Path Planning****	**Rule-Based Navigation**	**Integrated Proposed System (DQN-Based DRL)**
Path Planning Efficiency	25%	High	Medium	High
Adaptability to Environment Changes	25%	Medium	Low	High
Obstacle Avoidance	20%	High	Medium	High
Ability to Learn and Improve	15%	Low	Low	High
Collision Reduction	10%	High	Medium	High
Implementation Complexity	5%	Medium	Low	High
Overall Suitability	**100%**	**High**	**Medium**	**High**

The DQN-based proposed system is selected because it is designed to learn navigation decisions through interaction with the warehouse environment and aims to adapt to changing obstacles while generating efficient and collision-free paths.

3.5 Selected Design & Detailed Design

3.5.1 Selected Design

Based on the evaluation of alternative solutions, the DQN-Based Deep Reinforcement Learning approach was selected for the proposed system. The selection is based on its ability to enable the autonomous warehouse robot to learn navigation decisions through repeated interaction with the warehouse environment. Unlike fixed rule-based navigation, the DQN agent receives the current state of the environment, selects an action, and receives a reward or penalty based on the outcome. Through multiple training episodes, the robot gradually learns an improved navigation policy.

The selected design focuses on efficient and adaptive navigation from a source location to a destination while avoiding obstacles. The system aims to reduce unnecessary movement, path length, travel time, and collisions. The final design consists of two interconnected modules: Warehouse Environment & State Management and DQN-Based Adaptive Path Planning.

3.5.2 Warehouse Environment & State Management

The first module creates a two-dimensional warehouse environment containing the autonomous robot, destination, shelves, and obstacles. The environment provides the information required by the reinforcement learning agent to make navigation decisions. The current state of the robot includes its position, target location, and information about nearby obstacles. The robot can perform actions such as Up, Down, Left, Right, and Stop. After each action, the environment updates the robot's position and checks whether the robot has collided with an obstacle or successfully reached the destination. The resulting state and feedback are then provided to the DQN agent for further learning and decision-making.

Flow:

Warehouse Environment → Robot Observation → Current State → DQN Agent

3.5.3 DQN-Based Adaptive Path Planning

The second module is the main intelligence component of the proposed system. A **Deep Q-Network (DQN)** is used to estimate the value of the available navigation actions for the current robot state.

The DQN agent receives the state information and generates Q-values corresponding to the possible actions. Based on these values, the agent selects the most appropriate action. After executing the action, the robot receives feedback from the environment in the form of a reward or penalty.

The reward mechanism encourages the robot to move efficiently and successfully reach the destination. Penalties are applied when the robot collides with an obstacle, performs unnecessary movements, or follows a longer path. The system uses **Experience Replay** and a **Target Network** to improve training stability. The robot is trained over multiple episodes until it learns an effective navigation policy.

Flow:

State → DQN → Q-values → Best Action → Reward/Penalty → Learning → Improved Path

3.5.4 Detailed System Working

The detailed working of the proposed system begins with the creation of the warehouse environment. The robot is placed at a source location, and a destination is defined within the environment. Shelves and obstacles are also included to represent warehouse conditions. The robot observes its current state and sends this information to the DQN agent. The DQN evaluates the available actions and selects the most suitable movement. The selected action is performed in the warehouse environment, and the robot receives a reward or penalty based on the result.

If the robot moves efficiently towards the destination, it receives a positive reward. If it reaches the destination, an additional positive reward is provided. If a collision occurs or the robot makes unnecessary movements, a penalty is applied. This interaction is repeated during multiple training episodes, allowing the DQN agent to improve its navigation decisions and generate an adaptive and collision-free path.

3.5.5 System-Level Integration

The proposed system follows an integrated approach in which the warehouse environment and DQN agent continuously interact with each other. The environment provides the robot's state, including position, destination, and nearby obstacles. The DQN agent processes this state and selects an appropriate navigation action. The selected action changes the robot's position within the warehouse environment. The environment then evaluates the result and provides a corresponding reward or penalty. This feedback is used by the DQN agent to improve its learning. Therefore, the performance of the system depends on the continuous interaction between environment observation, action selection, reward evaluation, and learning.

The integrated design enables the robot to gradually develop an efficient navigation policy and supports the project's objective of adaptive, collision-free warehouse path planning.

3.5.6 Final Selected Architecture

The overall architecture of the proposed system can be represented as:

Warehouse Environment

↓

Robot Position + Destination + Obstacles

↓

State Representation

↓

DQN Agent

↓

Q-Value Calculation

↓

Action Selection

↓

Robot Movement

↓

Reward / Penalty

↓

Experience Replay & Learning

↓

Improved Navigation Path

This selected architecture supports the development of an adaptive warehouse robot path-planning system using Deep Reinforcement Learning.

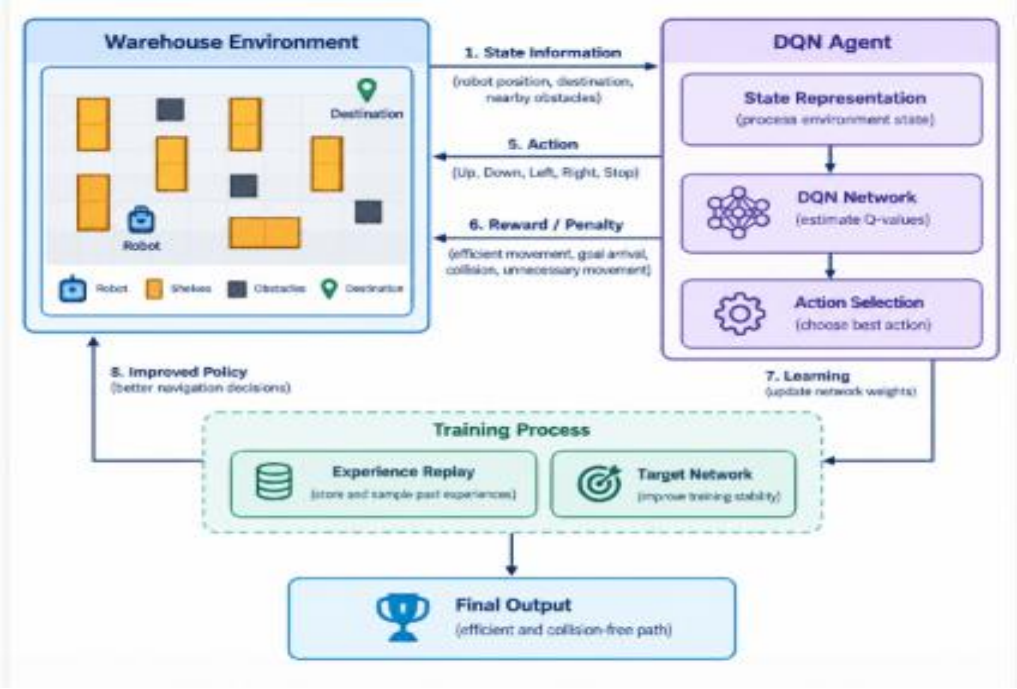


Figure 3.2 System Architecture of the DQN-Based Adaptive Path Planning for Autonomous Warehouse Robots

3.6 Trade-off Analysis

Design Decision	Alternative Considered	Benefit	Disadvantage	Final Decision & Justification
Path Planning Method	A* Path Planning	Efficient path generation using heuristic search	Limited adaptability when the environment changes	**DQN-Based DRL selected** because the proposed system aims to learn navigation decisions and adapt through interaction with the environment
Navigation Strategy	Rule-Based Navigation	Simple and easy to implement	Fixed rules cannot easily handle all possible navigation situations	**DQN-Based learning selected** to enable adaptive decision-making instead of relying only on predefined rules
Learning Approach	Traditional Fixed Path Planning	Does not require training	Requires predefined maps and may need repeated replanning	**Reinforcement Learning selected** because the robot can improve its navigation policy through repeated interaction

Action Selection	Random or Fixed Action Selection	Simple implementation	Can result in unnecessary movements and inefficient paths	**DQN action selection selected** to estimate Q-values and choose the most appropriate action
Training Method	Direct Training Without Experience Replay	Requires fewer components	Previous navigation experiences are not effectively reused	**Experience Replay selected** to use previous experiences and support improved learning
Network Training	Single Network Training	Simple architecture	Training can become less stable	**Target Network selected** to improve the stability of DQN training

3.7 Sustainability, Ethics, Safety, Risk & Security

Domain	Consideration	Evidence Evaluated	Impact on Decision
Sustainability	Reduce unnecessary robot movement and improve path efficiency	Path length, unnecessary movements, and completion time are considered as performance measures	The reward mechanism encourages efficient movement and shorter paths
Ethics	Ensure that system performance is evaluated fairly using measurable results	Success rate, path length, collision rate, completion time, and reward	System evaluation is based on measurable performance metrics rather than subjective judgement
Safety	Reduce the possibility of robot collisions with shelves and obstacles	Collision detection and collision penalties in the reward mechanism	Collision avoidance is included as a high-priority requirement in the DQN navigation system
Risk	Poor navigation decisions may result in inefficient paths or repeated collisions	Robot actions, rewards, penalties, collision rate, and navigation performance	Experience Replay, Target Network, and repeated training are used to improve learning stability and navigation performance
Security	Protecting communication and system data would be important in a real-world deployment	The current project is a simulation-based path-planning system and does not include a specific security mechanism	Security implementation is outside the current project scope and can be considered during future real-world deployment

Risk Register

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
Robot collision with shelves or obstacles	Medium	High	High	Implement collision detection and apply penalties for collision actions	Low
Generation of an inefficient or unnecessarily long path	Medium	Medium	Medium	Use rewards for efficient movement and penalties for unnecessary movements and longer paths	Low
Poor DQN training performance	Medium	High	High	Train the agent over multiple episodes and use Experience Replay and a Target Network	Medium
Failure to adapt effectively to changing obstacles	Medium	Medium	Medium	Train the agent using different environment states and obstacle conditions	Medium
Excessive completion time	Medium	Medium	Medium	Reward efficient movement and evaluate completion time during testing	Low
Simulation environment not fully representing a real warehouse	High	Medium	High	Clearly define simulation limitations and consider real sensor and robot integration as future work	Medium
Incorrect action selection during navigation	Medium	Medium	Medium	Use reward/penalty feedback to improve action selection during repeated training	Low
Unstable learning or inconsistent navigation performance	Medium	High	High	Use Experience Replay and Target Network to improve training stability	Medium

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

The project follows a Deep Reinforcement Learning-based development approach for adaptive warehouse robot path planning. Initially, a two-dimensional warehouse environment is created with a robot, destination, shelves, and obstacles. The environment provides the current state of the robot, including its position, target location, and nearby obstacles. A Deep Q-Network (DQN) agent receives this state and selects an appropriate navigation action. The environment then provides a reward or penalty based on the action outcome. Through repeated training episodes, Experience Replay and a Target Network are used to improve the learning stability and navigation policy of the robot.

Resources Used

Resource / Component	Purpose	Stage Used
Warehouse Environment	Simulates robot navigation, destination, shelves, and obstacles	Environment Development
Autonomous Robot Model	Represents the navigating warehouse robot	Simulation
Deep Q-Network (DQN)	Learns and selects suitable navigation actions	Model Development
Reward and Penalty Mechanism	Provides feedback for robot actions	Training
Experience Replay	Stores and reuses previous navigation experiences	Training
Target Network	Improves DQN training stability	Training
Performance Metrics	Measures navigation performance	Testing and Evaluation

4.2 Software Implementation & Modern Tools Used

The proposed system is implemented as a simulation-based Deep Reinforcement Learning application. The warehouse environment manages the robot position, destination, shelves, obstacles, movement actions, collision detection, and goal detection. The DQN agent receives the state information and estimates Q-values for the available actions. The action with the appropriate value is selected, and the

resulting movement is evaluated using a reward or penalty mechanism. The implementation process consists of creating the warehouse environment, defining the robot state and actions, developing the DQN model, implementing the reward mechanism, training the agent over multiple episodes, and evaluating the learned navigation performance.

Tool / Software / Equipment	**Purpose**	**Stage Used**
Python Programming Environment	Development of the warehouse simulation and learning logic	Implementation
Deep Reinforcement Learning Model	Adaptive path-planning decision-making	Model Development
DQN Algorithm	Estimation of action values and action selection	Training
Experience Replay	Storage and reuse of training experiences	Training
Target Network	Improved training stability	Training
Simulation Environment	Testing robot navigation and obstacle avoidance	Testing
Performance Metrics	Evaluation of success rate, path length, collision rate, completion time, and reward	Results Analysis

The DQN-based implementation and its supporting components are defined in the project design. However, the exact software libraries or simulation platform are not specified in the uploaded PPT and should only be added here if they were actually used in your implementation.

4.3 Test Plan & Verification

The test plan verifies whether the proposed warehouse robot navigation system meets its functional and performance requirements.

Requirement	**Test Method**	**Acceptance Criterion**
Robot should reach the destination	Run navigation episodes from source to destination	Robot successfully reaches the destination
Robot should avoid obstacles	Place obstacles along the navigation path	Robot avoids obstacles or collision is correctly detected
Robot should select valid actions	Monitor selected actions during navigation	Actions remain within the defined action set

System should reduce unnecessary movement	Measure the number of movements taken	Path does not contain excessive unnecessary movements
DQN should learn through training	Compare navigation behaviour over training episodes	Navigation performance shows improvement
Collision detection should function correctly	Intentionally move the robot towards an obstacle	Collision is detected and a penalty is applied
Goal detection should function correctly	Navigate the robot to the target location	Destination arrival is correctly identified

Verification Table

Specification	**Required Value**	**Achieved Value**	**Status (Pass/Fail)**
Destination Reached	Successful	To be measured	To be updated
Collision Avoidance	Collision reduction	To be measured	To be updated
Path Length	Efficient path	To be measured	To be updated
Completion Time	Reduced time	To be measured	To be updated
Success Rate	Improved success rate	To be measured	To be updated
Reward	Improved cumulative reward	To be measured	To be updated

The project identifies success rate, path length, collision rate, completion time, and reward as the primary evaluation measures.

4.4 Results

The proposed system is evaluated based on its ability to generate an efficient and collision-free path between the source and destination. The major performance parameters identified for evaluation are success rate, path length, collision rate, completion time, and accumulated reward.

The actual results obtained from the simulation should be presented using tables, graphs, and screenshots of the warehouse environment. The results can demonstrate the improvement in the robot's navigation behaviour as the DQN agent learns through multiple training episodes.

Suggested Results Table

Performance Metric	DQN-Based Proposed System	*A / Traditional Method**
Success Rate	To be measured	To be measured
Average Path Length	To be measured	To be measured
Collision Rate	To be measured	To be measured
Average Completion Time	To be measured	To be measured
Average Reward	To be measured	Not Applicable / To be measured

Suggested figures for this section:

- **Figure 4.1:** Warehouse Simulation Environment
- **Figure 4.2:** Robot Navigation Path Before Training
- **Figure 4.3:** Robot Navigation Path After Training
- **Figure 4.4:** Training Reward Across Episodes
- **Figure 4.5:** Comparison of DQN and A* Path Planning Performance

The PPT supports these performance measures but does not provide actual measured results.

4.5 Data Analysis & Engineering Judgment

The performance data should be analysed by comparing the navigation results obtained during different training stages. An improvement in success rate and accumulated reward would indicate that the DQN agent is learning more effective navigation behaviour. Similarly, a reduction in path length, collision rate, and completion time would indicate improved path-planning efficiency.

Engineering judgment should consider all performance measures together rather than relying on only one metric. For example, a shorter path may not be considered better if it results in a higher collision rate. Therefore, the final navigation policy should balance path efficiency, successful destination arrival, and collision avoidance.

Any deviation from expected results should also be analysed. Possible reasons may include insufficient training episodes, unsuitable reward values, complex obstacle arrangements, or limitations of the simulated environment. These observations can be used to improve the reward mechanism and training

process in future development. The proposed system is designed to evaluate navigation performance using multiple metrics, supporting engineering decisions based on overall system performance.

4.6 Comparison with Existing Solutions & Achievement of Objectives

Objective	Evidence	Achievement
Develop a warehouse environment	Two-dimensional environment with robot, destination, shelves, and obstacles	Fully / To be verified
Represent the robot state	Robot position, target location, and nearby obstacles are included	Fully / To be verified
Develop DQN-based path planning	DQN agent estimates action values and selects actions	Fully / To be verified
Avoid obstacles	Collision detection and penalties are included	To be verified through testing
Minimise path length	Efficient movement is encouraged through rewards	To be verified through results
Reduce collisions	Collision penalties are included	To be verified through results
Improve navigation adaptability	Agent learns through repeated interaction	To be verified through testing
Evaluate system performance	Success rate, path length, collision rate, completion time, and reward are defined	To be verified through experiments
Compare with A*	A* is identified as a conventional comparison approach	To be completed after testing

The proposed system differs from traditional fixed path-planning approaches by using a DQN agent that learns navigation decisions from environmental interaction and reward feedback.

4.7 Strengths & Limitations

Strengths

- Uses Deep Reinforcement Learning for adaptive navigation.
- Enables the robot to learn through interaction with the environment.
- Considers robot position, destination, and surrounding obstacles.

- Includes collision detection and destination detection.
- Uses a reward and penalty mechanism to guide navigation behaviour.
- Uses Experience Replay and a Target Network to support stable training.
- Evaluates multiple performance measures such as success rate, path length, collision rate, completion time, and reward.

Limitations

- The current project is based on a simulated two-dimensional warehouse environment.
- Actual real-world sensor integration is not included.
- Physical robot implementation is outside the current project scope.
- Performance depends on the quality of training and reward design.
- Multi-robot coordination is not included in the current design.
- Real-world warehouse conditions may introduce additional complexity.

These limitations are consistent with the future extensions identified in the project, including dynamic obstacles, multiple robots, real sensor data, and real-world robotic platforms.

4.8 Project Planning, Roles & Budget

Project Milestone Plan

Phase	Activity	Expected Outcome
Phase 1	Problem Identification and Literature Review	Understanding of warehouse path-planning approaches
Phase 2	Warehouse Environment Design	Two-dimensional simulated warehouse environment
Phase 3	State and Action Definition	Robot state and navigation actions established
Phase 4	DQN Model Development	DQN agent developed for action selection

Phase 5	Training	Agent trained over multiple episodes
Phase 6	Testing and Performance Evaluation	Navigation performance measured
Phase 7	Comparison and Analysis	Comparison with traditional approach
Phase 8	Report Preparation	Final Capstone Project Report

Team Roles and Contributions

Student	**Assigned Responsibility**	**Actual Contribution**
K. L. Khaviya Varshini	**Module 1 – Warehouse Environment and State Management**	Developed the warehouse environment, defined the robot position and destination, represented shelves and obstacles, and managed the robot state and navigation actions.
M. Nandhini	**Module 2 – DQN-Based Adaptive Path Planning**	Developed the DQN-based adaptive path-planning module, implemented the state-to-action decision process, designed the reward and penalty mechanism, trained the agent using Experience Replay and Target Network, and evaluated navigation performance.

The budget is an indicative academic prototype estimate rather than a purchase invoice. Actual component prices vary by supplier and quantity.

CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

The proposed Module 2 – DQN-Based Adaptive Path Planning focuses on enabling an autonomous warehouse robot to make intelligent navigation decisions using Deep Reinforcement Learning. The module is designed to process the current state of the warehouse environment and select appropriate actions for moving the robot towards its destination. The Deep Q-Network (DQN) approach enables the robot to learn from its interaction with the environment. The agent receives information about the robot position, destination, and surrounding obstacles and selects an appropriate navigation action. A reward and penalty mechanism is used to encourage efficient movement and successful destination arrival while discouraging collisions, unnecessary movements, and longer paths.

The module also uses Experience Replay and a Target Network to support stable learning during the training process. Through repeated training episodes, the DQN agent can improve its navigation policy based on previous experiences and environmental feedback.

Thus, Module 2 provides the intelligent decision-making component of the overall autonomous warehouse robot system. It supports the generation of an adaptive navigation path and contributes to reducing unnecessary movement and improving obstacle avoidance.

5.2 Achievement of Module Objectives

Module Objective	Achievement
Develop a DQN-based adaptive path-planning approach	Achieved through the design of the Deep Q-Network decision-making process
Process the robot's current state	Achieved by using environment state information as input to the DQN agent
Select suitable navigation actions	Achieved using Q-value estimation for the available actions
Implement reward-based learning	Achieved using rewards for efficient movement and destination arrival
Penalise undesirable navigation behaviour	Achieved through penalties for collisions and unnecessary movements
Improve learning stability	Achieved using Experience Replay and a Target Network

Support adaptive navigation	Achieved by allowing the agent to learn from repeated interaction with the environment
Evaluate navigation performance	Performance can be evaluated using success rate, path length, collision rate, completion time, and reward

5.3 Future Work

The current Module 2 implementation can be further improved in several ways. Future development can focus on extending the DQN-based path-planning system to handle more complex and realistic warehouse environments.

Possible future enhancements include:

- Introducing dynamic obstacles that move during robot navigation.
- Extending the system to support multiple autonomous robots.
- Integrating real-time sensor data for improved environment awareness.
- Implementing the system on a physical warehouse robot platform.
- Testing the system in a larger and more complex warehouse environment.
- Improving the reward mechanism for better navigation performance.
- Comparing the DQN approach with other advanced Deep Reinforcement Learning algorithms.

These future improvements can help transform the simulation-based Module 2 into a more practical and scalable autonomous warehouse navigation solution.

5.4 Final Outcome

The final outcome of Module 2 is the design and development of a DQN-Based Adaptive Path Planning mechanism for autonomous warehouse robots. The module provides an intelligent approach for selecting navigation actions based on the current state of the environment. The expected output of the module is an improved navigation policy that enables the robot to move towards its destination while considering obstacles and movement efficiency. The reward-based learning mechanism allows the DQN agent to improve its decisions through repeated training.

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APPENDICES

Appendix A

The objective of Module 2 is to develop an intelligent path-planning mechanism using a **Deep Q-Network (DQN)**. The module enables an autonomous warehouse robot to learn suitable navigation actions based on its current state, destination location, and surrounding obstacles.

The system uses a reward-based learning mechanism to encourage efficient movement and successful destination arrival while penalising collisions, unnecessary movements, and longer paths.

APPENDIX B – SYSTEM INPUTS AND OUTPUTS

Input	**Description**
Robot Position	Current location of the autonomous robot
Destination Position	Target location to be reached by the robot
Obstacle Information	Location of shelves and surrounding obstacles
Current State	Combined information about the robot and warehouse environment
Reward / Penalty	Feedback received after performing an action

Phase	**Activity**
Output	**Description**
Selected Action	Navigation action selected by the DQN agent
Updated Robot Position	New robot location after performing the action
Navigation Path	Path generated towards the destination
Reward Value	Feedback received for the selected action
Training Performance	Improvement in navigation behaviour over training episodes

APPENDIX C – ROBOT ACTION SPACE

The robot can perform the following navigation actions:

Action	**Description**
Up	Moves the robot upwards in the warehouse environment
Down	Moves the robot downwards

Left	Moves the robot towards the left
Right	Moves the robot towards the right
Stop	Stops the robot movement

APPENDIX D – REWARD AND PENALTY MECHANISM

The reward mechanism is used to guide the learning process of the DQN agent.

Robot Behaviour	**Feedback**
Efficient movement towards destination	Positive Reward
Successfully reaching the destination	High Positive Reward
Collision with an obstacle	Negative Reward / Penalty
Unnecessary movement	Negative Reward / Penalty
Longer navigation path	Negative Reward / Penalty

The reward and penalty mechanism helps the robot learn which actions produce better navigation outcomes.

APPENDIX E – PERFORMANCE METRICS

The following performance metrics are used to evaluate Module 2:

Performance Metric	**Purpose**
Success Rate	Measures how often the robot successfully reaches the destination
Path Length	Measures the total distance or number of movements required
Collision Rate	Measures the frequency of collisions with obstacles
Completion Time	Measures the time required to reach the destination

APPENDIX F – MODULE 2 WORKFLOW

Start
↓
Receive Current Warehouse State
↓
Provide State to DQN Agent
↓
Calculate Q-Values for Available Actions
↓
Select Best Action
↓
Move Robot in Warehouse Environment
↓
Check for Collision or Destination Arrival

↓

Provide Reward / Penalty

↓

Store Experience in Experience Replay

↓

Update DQN Model

↓

Repeat Until Destination is Reached or Episode Ends

APPENDIX G – ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
RL	Reinforcement Learning
DRL	Deep Reinforcement Learning
DQN	Deep Q-Network
Q-Value	Quality Value of an Action
MDP	Markov Decision Process
A*	A-Star Path Planning Algorithm

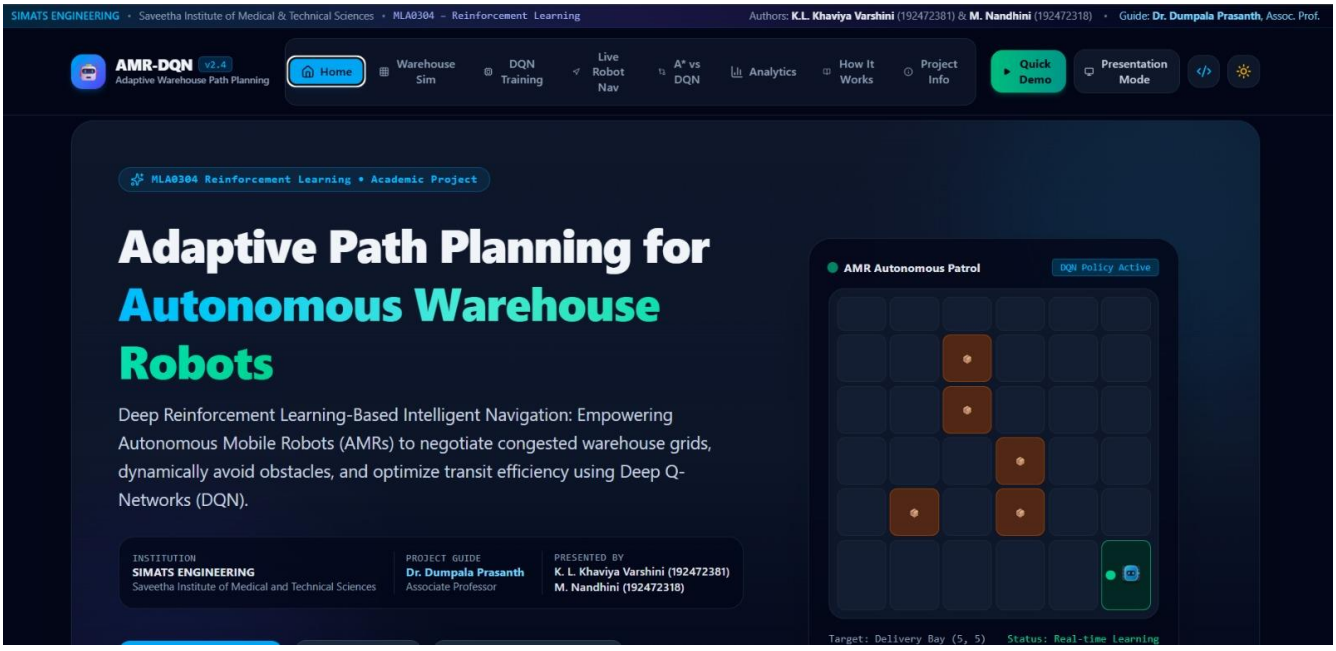


Fig 5.1 : Output